

This phrase means you are using a diagnostic tool (like `fttrace`, `ebpf`, or `gdb`) to monitor a specific network function inside the standard C library (`libc`) and tracing the execution path that led to it while running the `ping` command. [1]

Here is the breakdown of what each component means and how they work together:

The Core Components

- **probe:** Setting up a breakpoint or instrumentation point to intercept a function call when it executes.
- **libc:** The standard C library. It provides the core system calls and basic functions (like string handling and input/output) that almost all Linux programs use.
- **inet_pton:** A specific `libc` function. It stands for "Internet Protocol network address translation" (Presentation to Network). It converts a human-readable IP address string (like "192.168.1.1") into a binary format the system networking code understands.
- **backtrace it:** Capturing the "call stack" at the exact moment the function is triggered. This shows the sequence of function calls (Function A called Function B, which called Function C) that led to the probe point.
- **with ping:** Running the standard network diagnostic utility `ping` to trigger the entire chain of events.

How It Works in Practice

When you run a command like `ping google.com`, the following sequence occurs:

1. **Ping** starts up and takes the hostname or IP string you provided.
2. **Ping** calls `libc` functions to resolve and convert that string.
3. **inet_pton** executes to convert the IP string into raw bytes.
4. **Your probe** fires, pauses the execution instantly, and prints the stack trace. [1, 2, 3, 4]

Example Scenario (Using `bpftrace`)

If you run a system tracing tool with a command like `bpftrace -e 'uprobe:/lib/x86_64-linux-gnu/libc.so.6:inet_pton { print(ustack); }'`, and then run `ping 1.1.1.1` in another terminal, your tracing tool will output something like this:

text

```
inet_pton+0
main+542
__libc_start_main+243
_start+46
Use code with caution.
```

This output is the **backtrace**. It proves that the program started (`_start`), initialized (`__libc_start_main`), entered the main ping logic (`main`), and finally called `inet_pton`. Developers and system administrators use this technique to debug software bugs, analyze performance bottlenecks, or reverse-engineer how a program interacts with the operating system. [[1](#), [2](#), [3](#), [4](#), [5](#)]